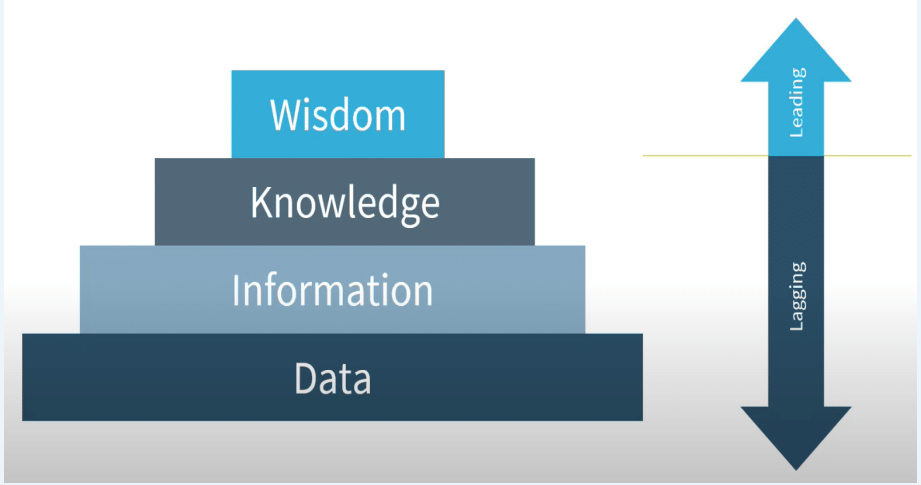
Statistics in Data Science

Dr Samira Fallahpour

the eminent statistician whose ideas developed over 50 years ago form the foundation of data science

john Tukey

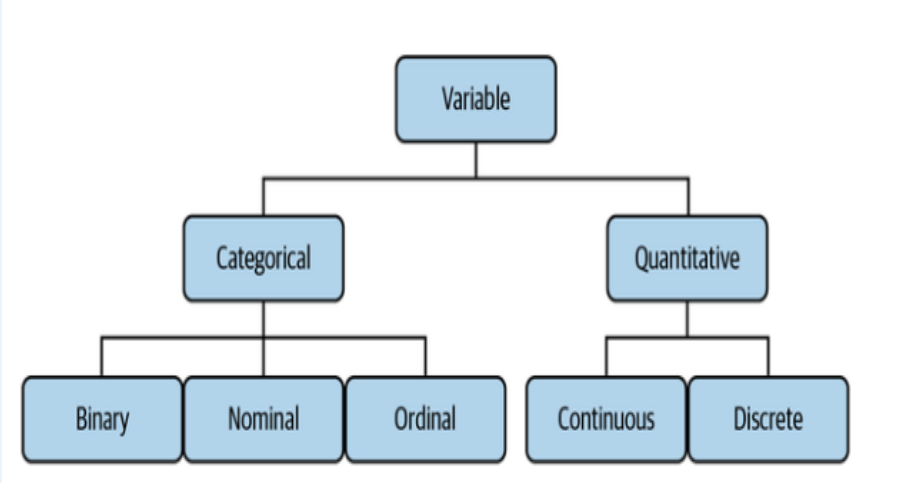
Data pyramid



The data pyramid, also known as the DIKW pyramid, is a hierarchical model that illustrates the progression from raw data to wisdom. This model is widely used in information science and knowledge management to understand how data are transformed into valuable insights and actionable wisdom.

Type of Variable

Quantitative: Data that are expressed numerically and measured on a numerical scale.
Continuous: Data that can take any value within a given range.
(Synonyms: interval, decimal, numerical — e.g., house price.)
Discrete: Data that consist only of whole-number values.
(Synonyms: integer values, counts — e.g., the number of students in a class.)



Qualitative: Data that consist only of specific values from a set representing different groups or categories.

(Synonym: categorical)

Binary (Dichotomous): A special type of categorical data that includes only two possible values, such as 0 or 1, true or false, female or male.

(Synonyms: dichotomous, logical, Boolean)

Ordinal (Ranked): A special type of categorical data that have a meaningful order. For example, level of salary paid to employees (low, medium, high), level of education (university, high school, etc.), level of cost, level of risk, customer/social class categories (high, medium, etc.).

Nominal: A special type of categorical data that consist of unique values without any inherent order. Examples include assigning numbers to industries, equipment, provinces and countries, blood types, colors, field of study, and place of birth.

Interval: A special type of data with interval values. In interval data, not only are groups, classifications, and their order specified, but the magnitude of differences between groups is also measurable. Examples include weather degree, temperature, hearing level, intelligence level, and language test scores.

Nominal Data:

Application and Description: Used for classifying objects, individuals, or characteristics.

Examples of Nominal Data: Assigning numbers to different industries, equipment, provinces and countries, blood types, colors, field of study, place of birth, gender.

Common Analytical Approach: Using percentages and frequency statistics.

Strengths and Weaknesses:

Lacks quantitative interpretive power; only classifies attributes.

The weakest scale among measurement types.

In this scale, only differences are identified; the magnitude or degree of differences cannot be measured.

Nominal Data:

aplication and Description: In addition to having nominal characteristics, ordinal data also possess an inherent order, where ranking is important.

Examples of Ordinal Data: Level of education (university, high school, etc.), income level, cost level, risk level, customer/social class categories (high, medium, etc.).

Common Analytical Approach: Spearman correlation coefficient.

Strengths and Weaknesses (Ordinal Scale):

•In this scale, there is no true basis for assigning absolute values.
•Therefore, a more advanced scale, called the interval scale, is required.
•Ranking is possible, but the exact distance between ranks cannot be determined.

Interval Data:
Application and Description:
In this type of scale, in addition to having ordinal characteristics, the distance between categories is expressed numerically.
Examples of Interval Data:
Language test scores, intelligence test scores, temperature scales, temperature intervals, hearing level.
Common Analytical Approach:
All descriptive and inferential statistics can be applied, such as mean, median, mode, and frequency.
Strengths and Weaknesses:
•Used for equal intervals that are meaningful for addition and subtraction; however, the zero point is arbitrary.
•Negative numbers can also be meaningful in interval data.
•In addition to possessing the features of nominal and ordinal scales, it can measure and quantify the differences between values.
Note:
In other words, in language test scores (which are considered interval data), the difference between scores of 90 and 100 is 10 units, which is exactly equal to the difference between scores of 70 and 80 on the same scale.

Ratio Scale:

Application and Description:

This scale represents the highest level of measurement and is the most quantitative scale. In addition to possessing all the characteristics of the three previous scales (nominal, ordinal, and interval), it has a unique feature: a true zero point (absolute zero).

Examples of Ratio Data:

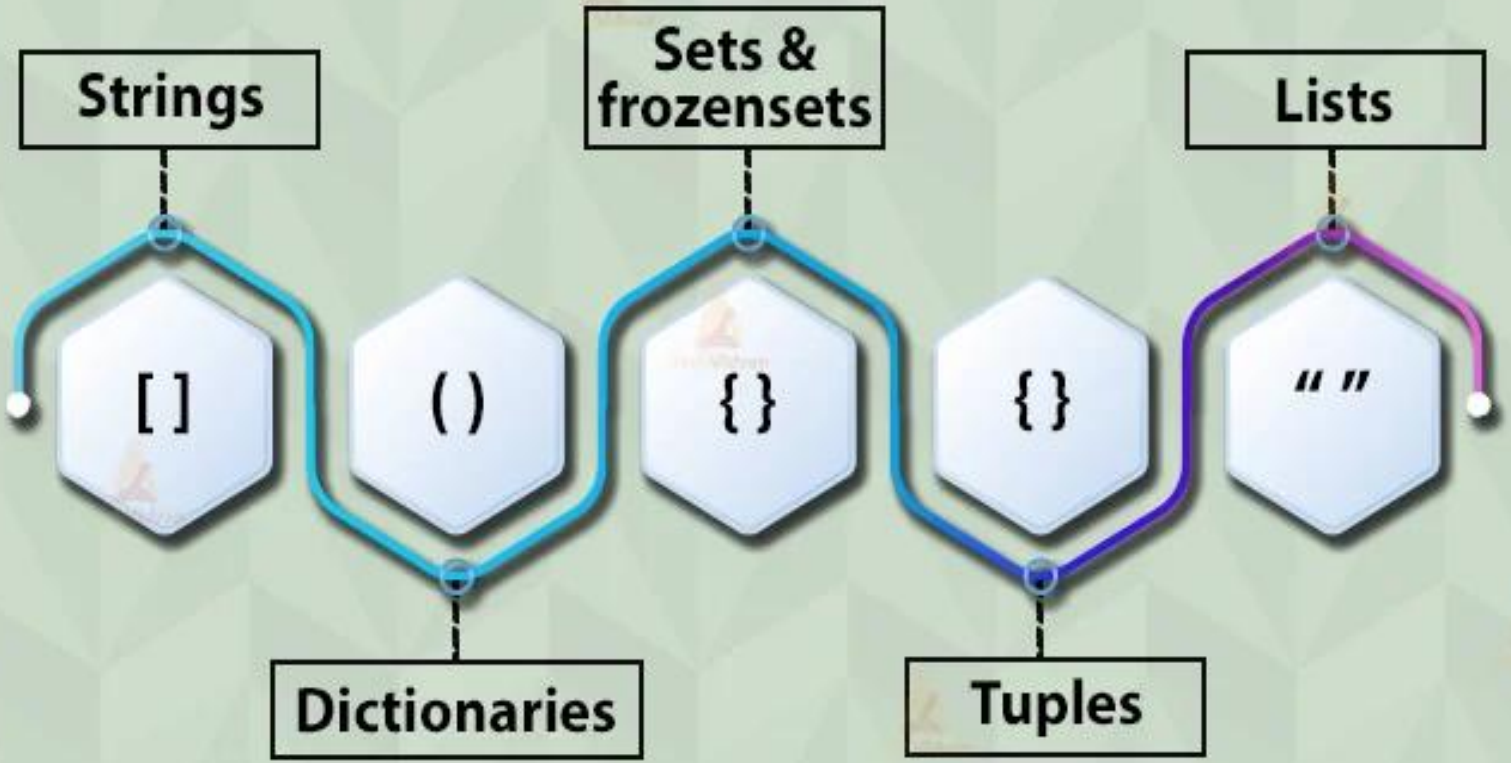
Birth rate, death rate, cholesterol level, divorce rate, age, weight, height, number of individuals, city population, number of years of education.

Strengths and Weaknesses:

•In this scale, all arithmetic operations—addition, subtraction, multiplication, and division—can be meaningfully applied.

•For example, units such as pounds and grams have a true zero point, making ratio comparisons meaningful.

Python Data Structures



TechVidvan

Data Structure	Dimensionality	Format	View
Series	1D	Column	
DataFrame	2D	Single Sheet	
Panel	3D	Multiple Sheets	

Series

	name
0	Rukshan
1	Prasadi
2	Gihan
3	Hansana

	age
0	25
1	25
2	26
3	24

	marks
0	85
1	90
2	70
3	80

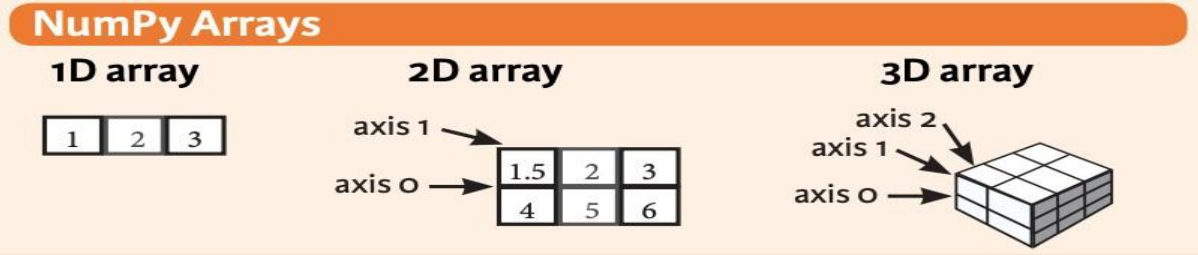
DataFrame

	name	age	marks
0	Rukshan	25	85
1	Prasadi	25	90
2	Gihan	26	70
3	Hansana	24	80

Panel

	name	age	marks
0	Rukshan	25	85
1	Prasadi	25	90
2	Gihan	26	70
3	Hansana	24	80

Image Copyright: Rukshan Pramoditha



Series

Definition: A one-dimensional data structure that stores data along with their corresponding index labels.
Similarity: It is similar to a list or an array, but with the ability to assign a label to each element.
Usage: It is used for storing and processing simple one-dimensional data

```
import pandas as pd
data = pd.Series ( [25, 30, 22], index = [ ‘sara’ , ‘ali’ , ‘reza’])
```

DataFrame

Definition: A two-dimensional data structure that stores data in rows and columns (like a table).
Similarity: It is similar to an Excel spreadsheet or a table in a SQL database.
Usage: It is used for storing and processing more complex and multidimensional data.

```
import pandas as pd

data = pd.DataFrame({
    'Name': ['Ali', 'Reza', 'Sara'],
    'Age': [25, 30, 22],
    'Job': ['Engineer', 'Doctor', 'Student']
})
```

Panel

Definition: A three-dimensional data structure that stores data as multiple DataFrame objects.
Similarity: It is like a data cube composed of multiple DataFrames.
Usage: It is used for storing and processing more complex data that require three dimensions (such as economic or time-based data).

```
import pandas as pd

data = {
    'Country1': pd.DataFrame({'Year': [2020, 2021], 'Production': [100, 120]}),
    'Country2': pd.DataFrame({'Year': [2020, 2021], 'Production': [150, 160]})
}

panel = pd.Panel(data)
```

Scalars, Vectors and Matrices

And when we include matrices we get this interesting pattern:

- A **scalar** is a number, like **3, -5, 0.368, etc**,
- A **vector** is a **list** of numbers (can be in a row or column),
- A **matrix** is an **array** of numbers (one or more rows, one or more columns).

Scalar	Vector	Matrix
24 →	$\begin{bmatrix} 2 & -8 & 7 \end{bmatrix}$ row	$\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}$
	or column $\begin{bmatrix} 2 \\ -8 \\ 7 \end{bmatrix}$	row(s) × column(s)

In fact a **vector is also a matrix**! Because a matrix can have just one row or one column.

So the rules that work for matrices also work for vectors.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 7 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}$$

A Matrix

(This one has 2 Rows and 3 Columns)

2x4=8

$$2 \times \begin{bmatrix} 4 & 0 \\ 1 & -9 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 2 & -18 \end{bmatrix}$$

"Dot Product"

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & \\ & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ & \end{bmatrix}$$

Notes	Python numpy.ndarray

We have a [5 x 3] matrix.	`>>> W`
	
	`array([[3, 1, 7],`
	`[7, 1, 4],`
	`[8, 6, 1],`
	`[8, 5, 1],`
	`[4, 3, 8]])`

Display the first row.	`>>> W[0, :]`
	
	`array([3, 1, 7])`

Display the size of the first row.	`>>> W[0, :].shape`
	
	`(3,)`

Display the first column.	`>>> W[:, 0]`
	
	`array([3, 7, 8, 8, 4])`

Display the size of the first column.	`>>> W[:, 0].shape`
	
	`(5,)`

Notes	Python numpy.ndarray

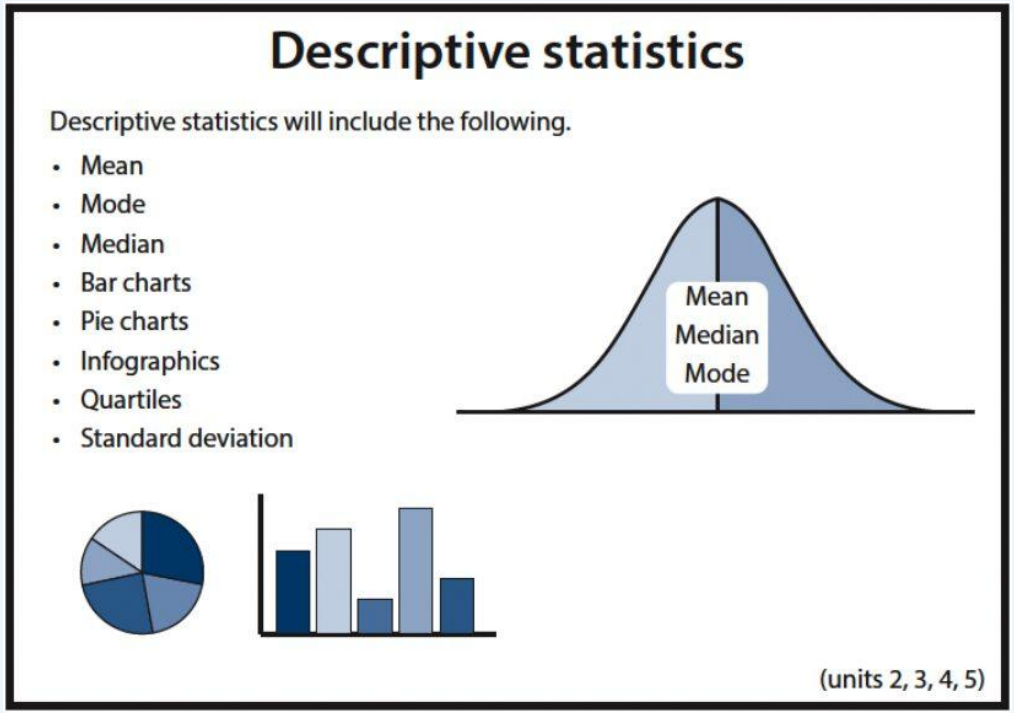
We have a [5 x 3] matrix.	`>>> W`
	
	`array([[3, 1, 7],`
	`[7, 1, 4],`
	`[8, 6, 1],`
	`[8, 5, 1],`
	`[4, 3, 8]])`

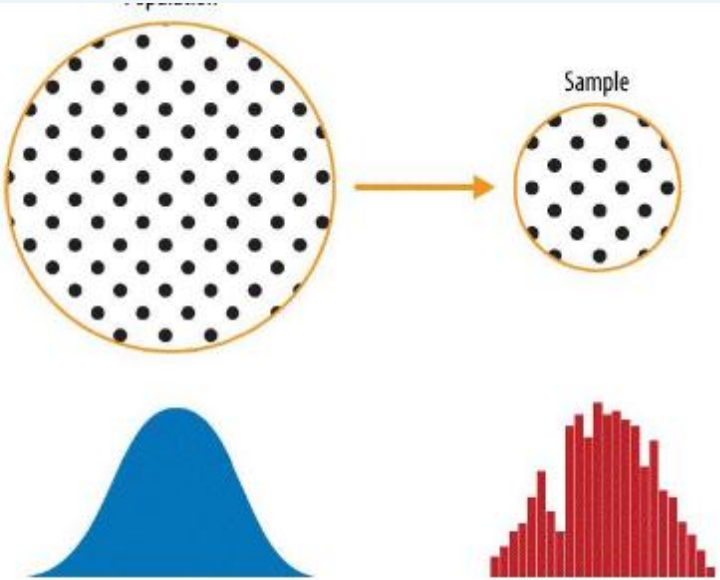
And a vector with 3 values.	`>>> x`
	
	`array([1, 2, 3])`

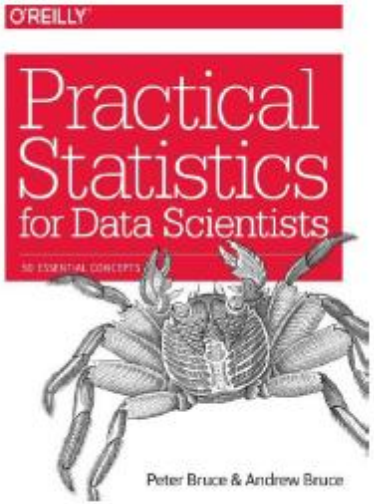
Multiply W and x with `*`	`>>> W * x`
	
	`array([[ 3, 2, 21],`
	`[ 7, 2, 12],`
	`[ 8, 12, 3],`
	`[ 8, 10, 3],`
	`[ 4, 6, 24]])`

Case	Age	Job	Marital	Education	Default	Balance	Loan	Duration	Campaign	Fraud
1	58	management	married	MSc	no	0	no	261	2	no
2	44	technician	single	Secondary	yes	29	no	151	1	yes
3	33	entrepreneur	married	BSc	no	2	yes	76	1	no
4	47	blue-collar	married	N.A	yes	0	no	92	1	no
5	33	unknown	single	Diploma	no	1	no	198	2	no
6	35	management	married	Tertiary	no	231	no	0.2	1	no
7	28	management	single	PhD	no	0	yes	217	1	N.A
8	42	unknown	divorced	Tertiary	yes	2	no	0	3	yes
9	58	unknown	married	Primary	no	121	no	50	1	no
10	115	unknown	single	Secondary	yes	593	no	55	1	no
11	41	admin.	divorced	BA	yes	270	no	222	2	no
12	29	admin.	single	BSc	no	390	no	137	1	no
13	53	technician	married	Diploma	no	6	no	517	2	no

In the world of data, analyzing and interpreting information plays a crucial role in scientific, business, and everyday decision-making. Raw data are usually presented as a collection of numbers, observations, or qualitative information, which are difficult to understand without proper processing and analysis.
Descriptive statistics is an important branch of statistics that provides tools for summarizing, describing, and presenting data. Charts and graphs help us visualize how data are distributed and enable us to better understand the relationships between the mean, median, and mode.







Sample Mean versus Population Mean

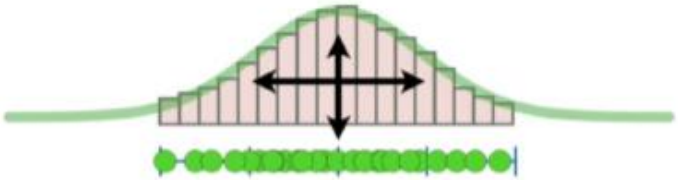
The symbol $\bar{x}$ (pronounced x-bar) is used to represent the mean of a sample from a population, whereas μ is used to represent the mean of a population. Why make the distinction? Information about samples is observed, and information about large populations is often inferred from smaller samples. Statisticians like to keep the two things separate in the symbology.

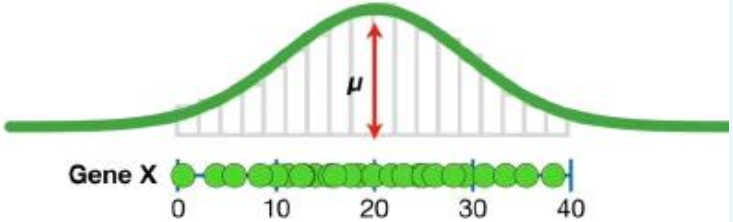
Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$	$\overline{X} = \frac{\sum_{i=1}^{n} x_i}{n}$
N = number of items in the population	n = number of items in the sample

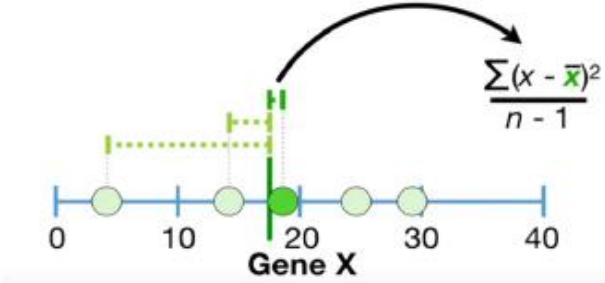
Population Variance	Sample Variance
$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$	$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$
σ^2 = population variance x_i = value of ith element μ = population mean N = population size	s^2 = sample variance x_i = value of ith element $\bar{x}$ = sample mean n = sample size

	SAMPLE	POPULATION
MEAN	$\bar{x}$	μ
STANDARD DEVIATION	s	σ
	STATISTIC	PARAMETER

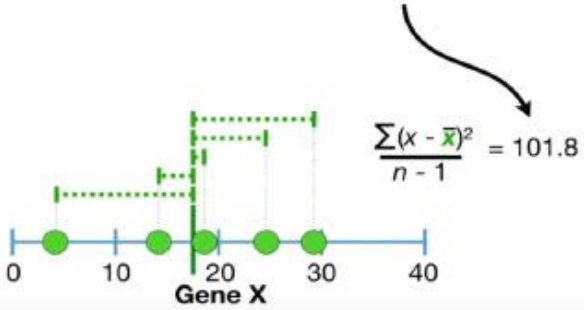
Statistics Fundamentals…

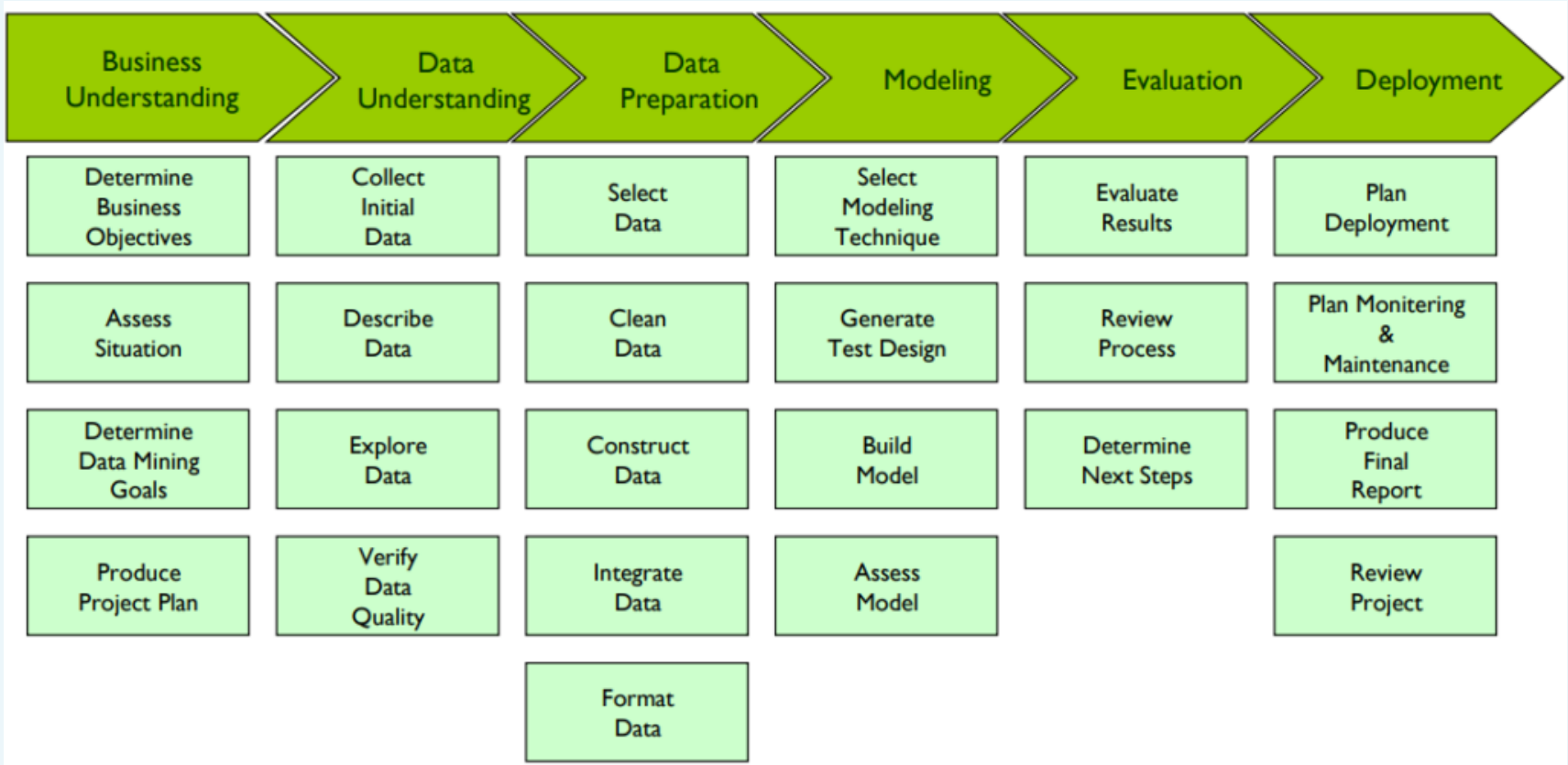






…and then we estimated the **variance**.




Business Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment
Determine Business Objectives
Assess Situation
Determine Data Mining Goals
Produce Project Plan
Collect Initial Data
Describe Data
Explore Data
Verify Data Quality
Select Data
Clean Data
Construct Data
Integrate Data
Format Data
Select Modeling Technique
Generate Test Design
Build Model
Assess Model
Evaluate Results
Review Process
Determine Next Steps
Plan Deployment
Plan Monitering & Maintenance
Produce Final Report
Review Project

A basic notion in probability theory is *random experiment* an experiment whose outcome cannot be determined in advance. The set of all possible outcomes of an experiment is called the *sample space* of that experiment, and we denote it by S.

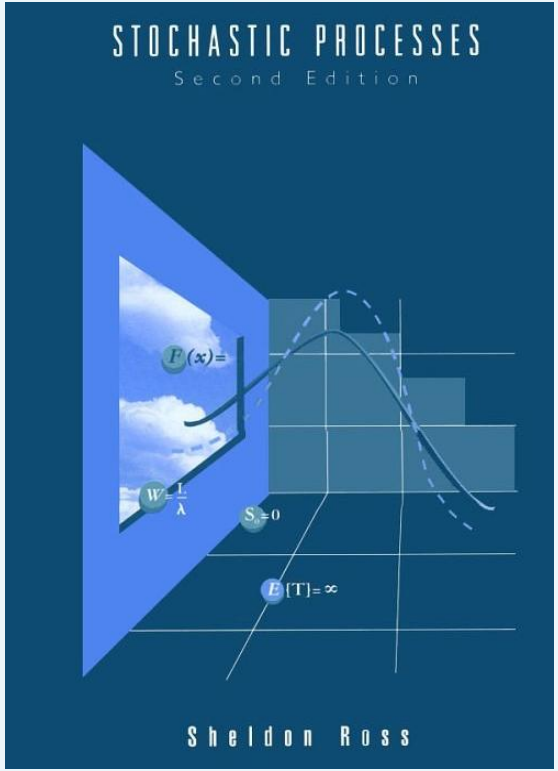
An *event* is a subset of a sample space, and is said to occur if the outcome of the experiment is an element of that subset. We shall suppose that for each event E of the sample space S a number $P(E)$ is defined and satisfies the following three axioms*:

Axiom (1) $0 \leq P(E) \leq 1$.

Axiom (2) $P(S) = 1$

Axiom (3) For any sequence of events $E_1, E_2, \ldots$ that are mutually exclusive, that is, events for which $E_iE_j = \phi$ when $i \neq j$ (where ϕ is the null set),

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$



1.1.1. If $E \subset F$, then $P(E) \leq P(F)$.

1.1.2. $P(E^c) = 1 - P(E)$ where E^c is the complement of E.

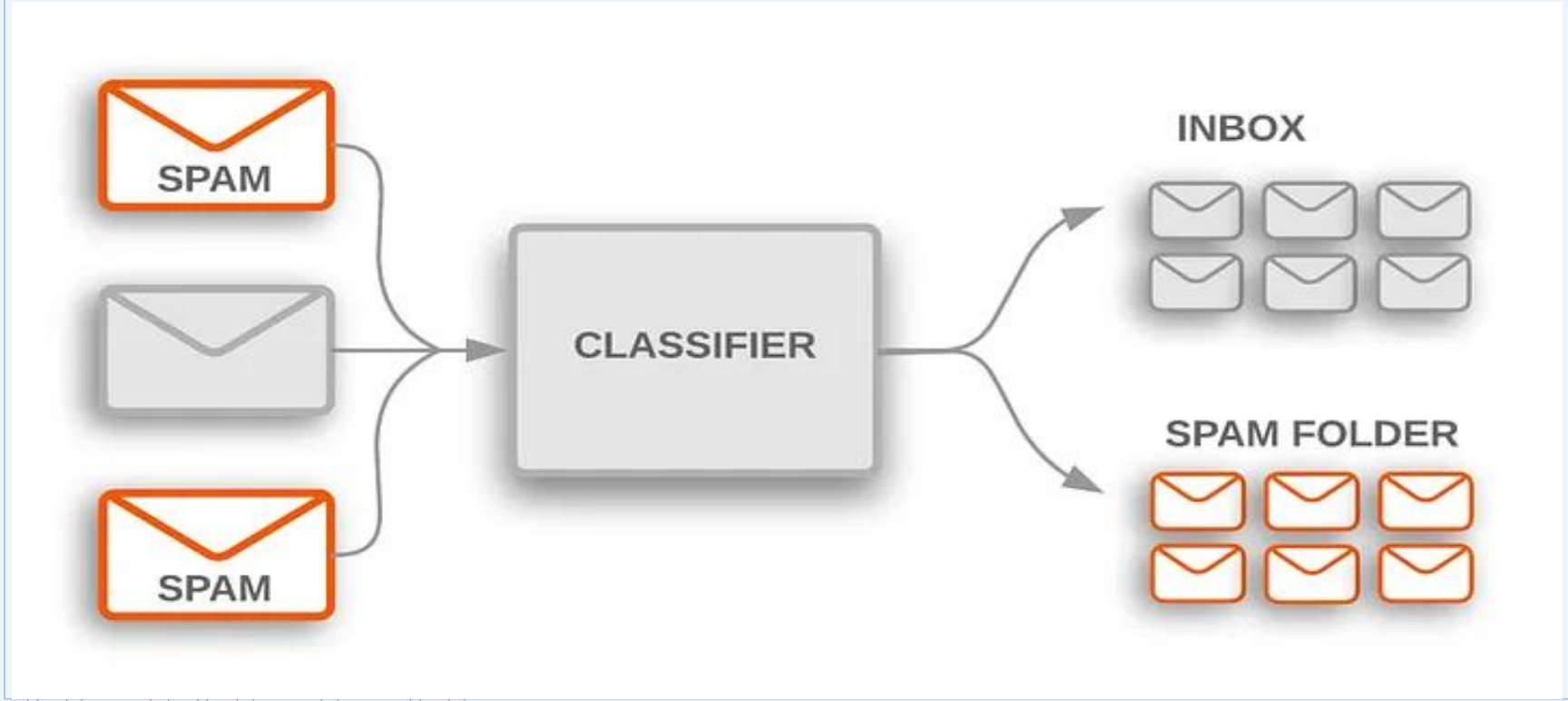
1.1.3. $P(\bigcup_1^n E_i) = \sum_1^n P(E_i)$ when the E_i are mutually exclusive.

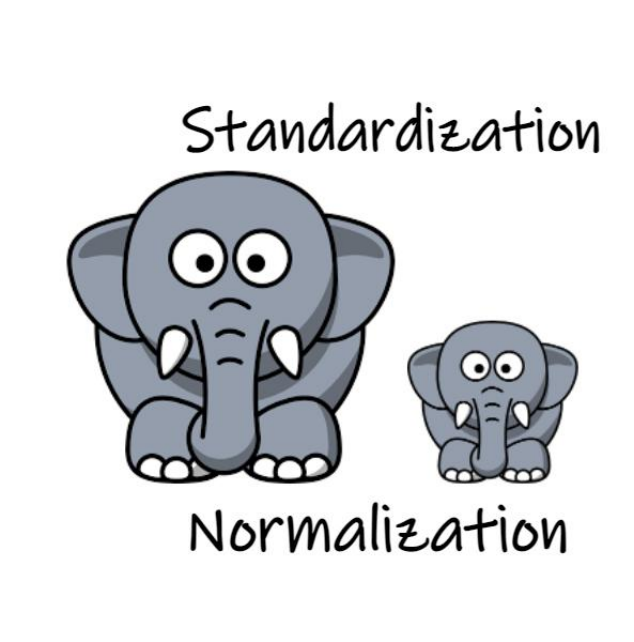
1.1.4. $P(\bigcup_1^\infty E_i) \leq \sum_1^\infty P(E_i)$.

The inequality (1 1 4) is known as *Boole's inequality*

* Actually $P(E)$ will only be defined for the so-called measurable events of S But this restriction need not concern us

Bayes' Theorem in email spam filtering





Standardization

Also Called ***Z-score normalization***

$$z = \frac{x - \text{mean}(x)}{\text{std}(x)}$$

Standardizing the features so that they are **centered around 0 with a standard deviation of 1** is not only important if we are **comparing measurements that have different units**, but it is also a general **requirement for many machine learning algorithms.**

Normalization

Also Called ***Min-Max scaling***

$$X_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

The data is scaled to a **fixed range(usually 0 to 1)**.The cost of having this bounded range - in contrast to standardization - is that we will end up with smaller standard deviations, which can **suppress the effect of outliers.**

Normalization or Standardization

- Feature Scaling means scaling features to the same scale.

- Normalization scales features between 0 and 1, retaining their proportional range to each other.

Normalization

new value → $X' = \dfrac{x - \min(x)}{\max(x) - \min(x)}$ ← original value

- Standardization scales features to have a mean (u) of 0 and standard deviation (a) of 1.

Standardization

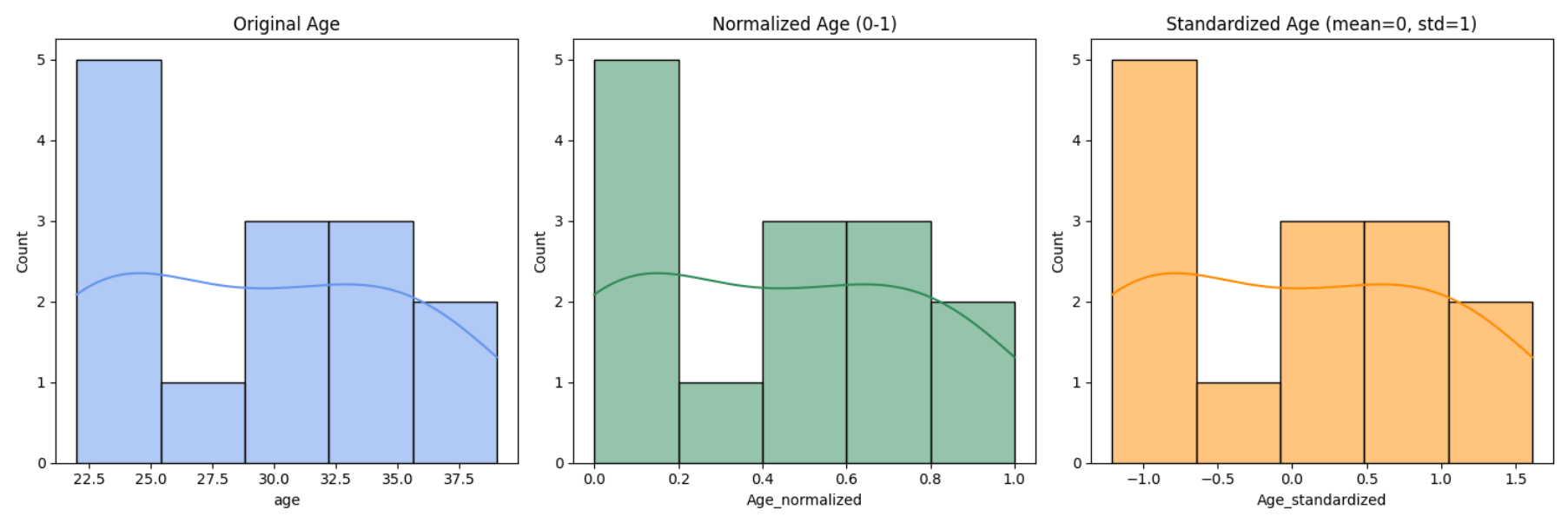
new value → $X' = \dfrac{x - u}{a}$ ← mean, ← standard deviation

original value

```python
import pandas as pd

df = pd.read_excel('dataset.xlsx')
df['Age_normalized'] = (df['age'] - df['age'].min()) / (df['age'].max() - df['age'].min())
df['Age_standardized'] = (df['age'] - df['age'].mean()) / df['age'].std()
df['Salery_normalized'] = (df['salery'] - df['salery'].min()) / (df['salery'].max() -
df['salery'].min())
df['Salery_standardized'] = (df['salery'] - df['salery'].mean()) / df['salery'].std()

print(df.head(15))
print("Age min", df['age'].min())
print("Salery min", df['salery'].min())
print("Age max", df['age'].max())
print("Salery max", df['salery'].max())
print(df[['age', 'Age_normalized', 'Age_standardized']].head())
print(df[['salery', 'Salery_normalized', 'Salery_standardized']].head())
df.to_excel("output_scaled.xlsx", index=False)
```

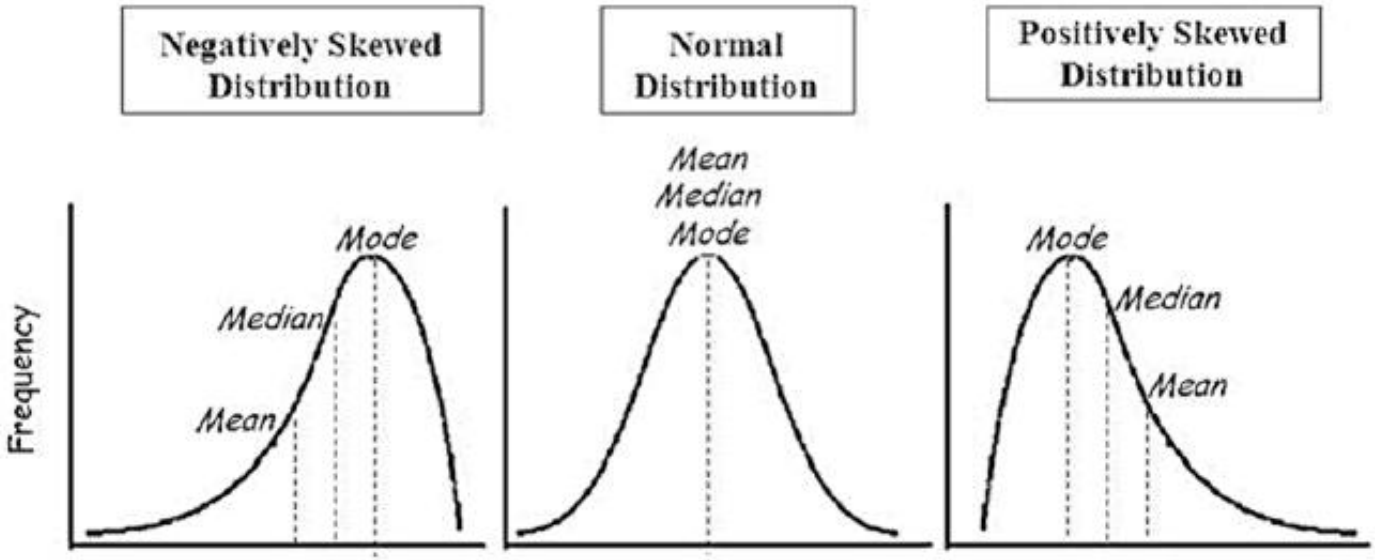

Original Age
Count
age
22.5
25.0
27.5
30.0
32.5
35.0
37.5
Normalized Age (0-1)
Count
Age_normalized
0.0
0.2
0.4
0.6
0.8
1.0
Standardized Age (mean=0, std=1)
Count
Age_standardized
−1.0
−0.5
0.0
0.5
1.0
1.5

Difference between Normalisation and Standardisation

S.NO.	NORMALISATION	STANDARDISATION
1.	Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
2.	It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
3.	Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
4.	It is really affected by outliers.	It is much less affected by outliers.
5.	Scikit-Learn provides a transformer called `MinMaxScaler` for Normalization.	Scikit-Learn provides a transformer called `StandardScaler` for standardization.
6.	This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of original data to the origin and squishes or expands.
7.	It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
8.	It is a often called as Scaling Normalization	It is a often called as Z-Score Normalization.

Central Tendency & Normal Distributions

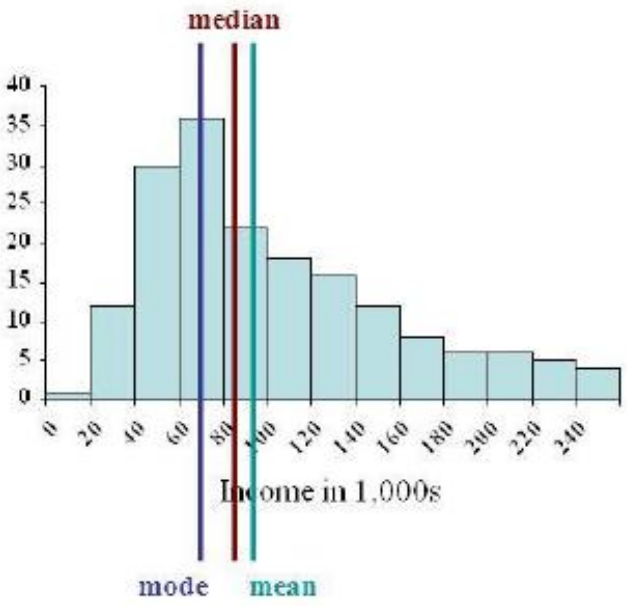
- The mean and the median are affected by skewness, or lack of symmetry in the data.



Skewed distributions

- Skewness refers to the asymmetry of the distribution
- A positively skewed distribution is asymmetrical and points in the positive direction.

Mode = 70,000$
Median = 88,700$
Mean = 93,600$



•**mode** < **median** < **mean**

Outliers Detection – Box Plot

FIVE NUMBER SUMMARY

MINIMUM	1ST QUARTILE	MEDIAN	3RD QUARTILE	MAXIMUM
	25	33	36	

10 11 12 25 25 27 31 33 34 34 35 36 43 50 59

$$\text{POSITION OF MEDIAN} = \frac{n+1}{2}$$

$$\text{POSITION OF Q1} = \frac{n+1}{2}$$

$$\text{POSITION OF Q3} = \frac{n+1}{2}$$